Calibrating the laser camera

A step-by-step guide for teachers, with pictures of each screen

What this is for. Pupils line their design up on the live camera picture at the laser PC. For that to land where the laser will really cut, the Editor has to know how the camera picture maps onto the bed. You teach it once per laser: the laser cuts a few squares of a known size at a known place, and you click their corners in the picture. Because the squares sit at known bed positions, this works even when the bed is much larger than the camera can see.

Before you start

- The laser's Cutter is connected and the Station page shows **Connected · Idle** with a camera picture available.
- A **teacher token** (or school admin token). You type it into the dialog; it is used for the calibration only and is never stored. Pupils cannot calibrate.
- A **scrap of material** at least 160 mm by 160 mm (for the default four 40 mm squares), placed where the camera can see it. Plywood is the default.
- Plan on about five minutes. The laser cuts for under a minute.

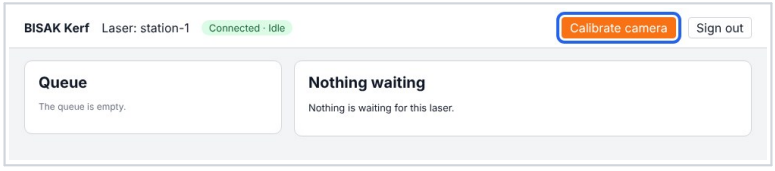
The five stages

- **1. Open it** from the Station page and enter a teacher token (steps 1 to 3).
- **2. Choose where the squares go** (step 4).
- **3. The laser cuts them**, tracing the outline first with the laser off (step 5).
- **4. Click every corner** in the camera picture, square by square (steps 6 to 8).
- **5. Save** (step 9). Pupils can then line up their designs (steps 10 to 13).

About the pictures. They were taken with the pretend laser and its test camera picture (a plain scrap on a 400 mm bed), so a real camera will look darker and more curved near the edges. Everything else is the real screen. The blue boxes were added to point at the thing to look at.

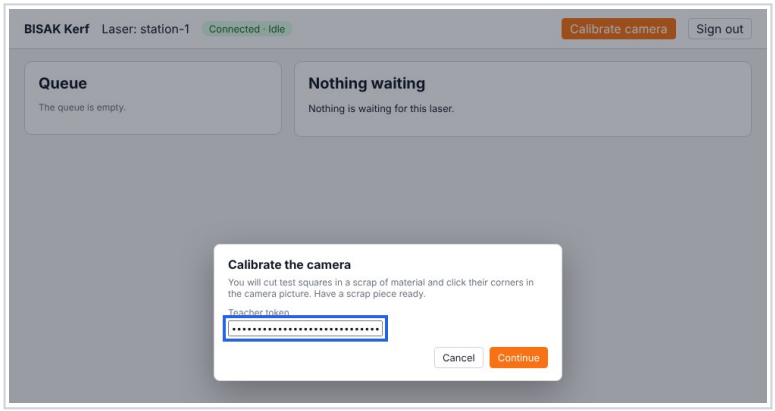
Part 1 · Calibrating (teacher, at the laser PC)

1 Open the Station page and press Calibrate camera



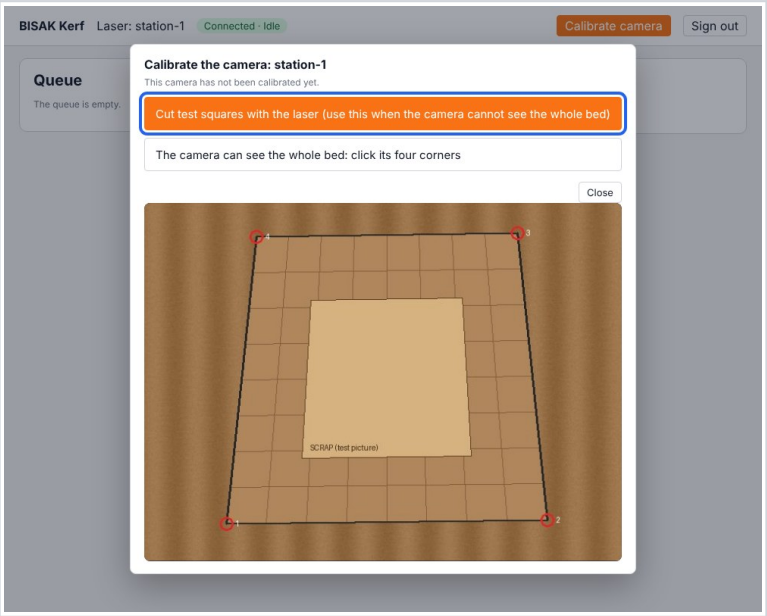
Sign in to the Editor on the laser PC with the station token. The button is in the top bar. It is orange and says **Calibrate camera** until the laser has a calibration; after that it says **Recalibrate camera**. It only appears while a camera picture is arriving.

2 Enter a teacher token



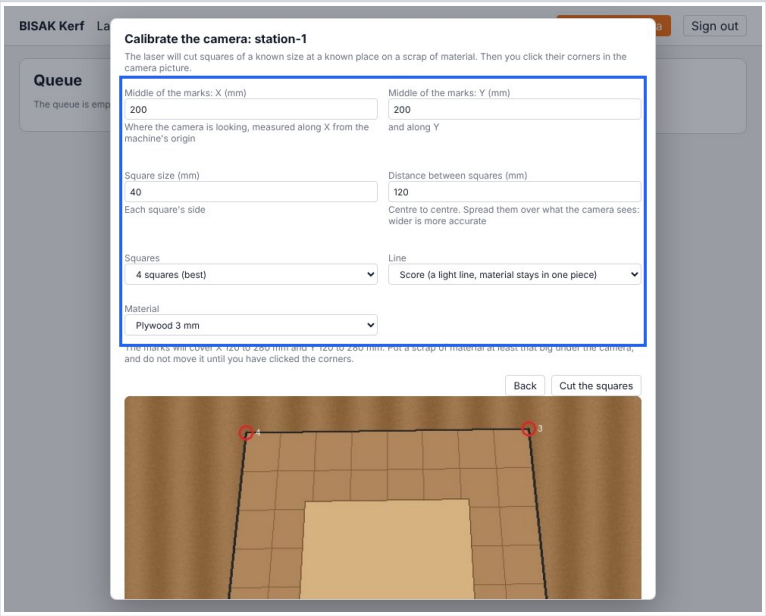
Calibrating changes what every pupil sees, so it asks for a teacher's token. It is checked, used for this dialog, and forgotten when the dialog closes. The station's own token cannot change a calibration.

3 Choose how to calibrate



Choose **Cut test squares with the laser**. Use it whenever the camera cannot see the whole bed, which is the usual case. The other choice, *click the bed's four corners*, is only for a camera that can see the entire bed.

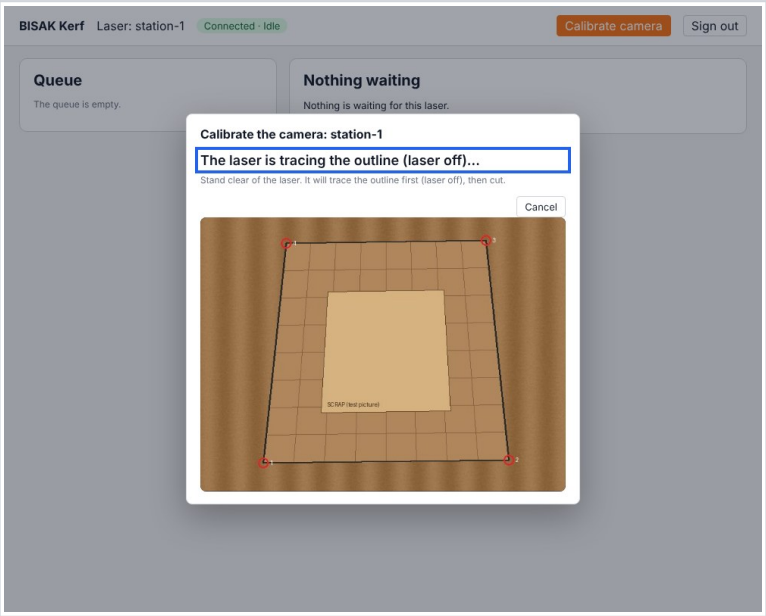
4 Decide where the squares go



These settings say where the laser will cut the squares:

- **Middle of the marks X and Y** (mm): the spot on the bed under the middle of the camera picture, measured from the machine's origin.
- **Square size** (default 40 mm) and **distance between squares** (default 120 mm, centre to centre). Spread them over what the camera sees: a wider spread is more accurate.
- **Squares**: 4 (best) or 1. **Line**: Score is a light line that leaves the scrap in one piece; Cut goes right through. **Material**: pick the scrap you have.
- The sentence under the form tells you the area the marks will cover. Put the scrap there and do not move it until you have finished clicking.

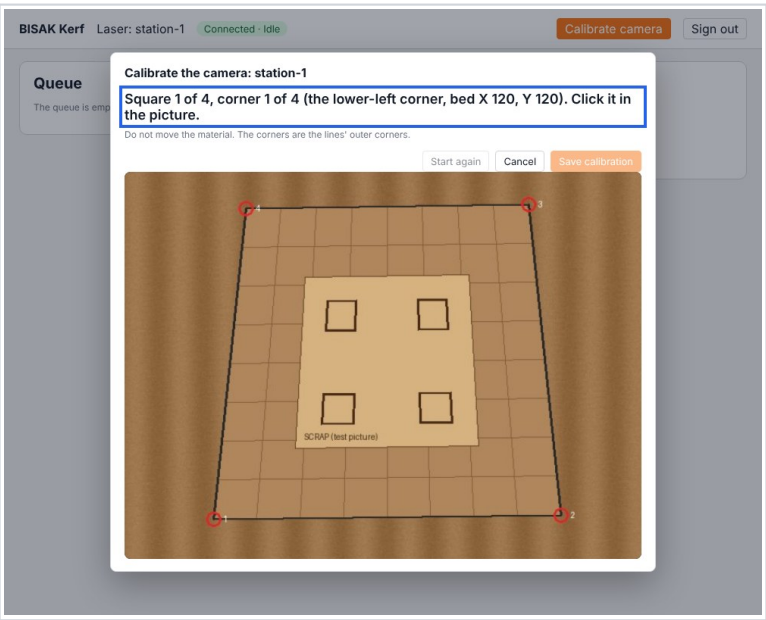
5 Press Cut the squares and stand clear



The dialog runs the job for you. First the laser **traces the outline with the laser off** (shown here), then it **cuts the squares**. Stand clear and keep the lid closed or the area supervised, as for any cut.

- If anything goes wrong (the laser stops, you cancel, a timeout) the calibration job is cancelled so it cannot sit in the pupils' queue, and you are taken back to the settings with the reason.

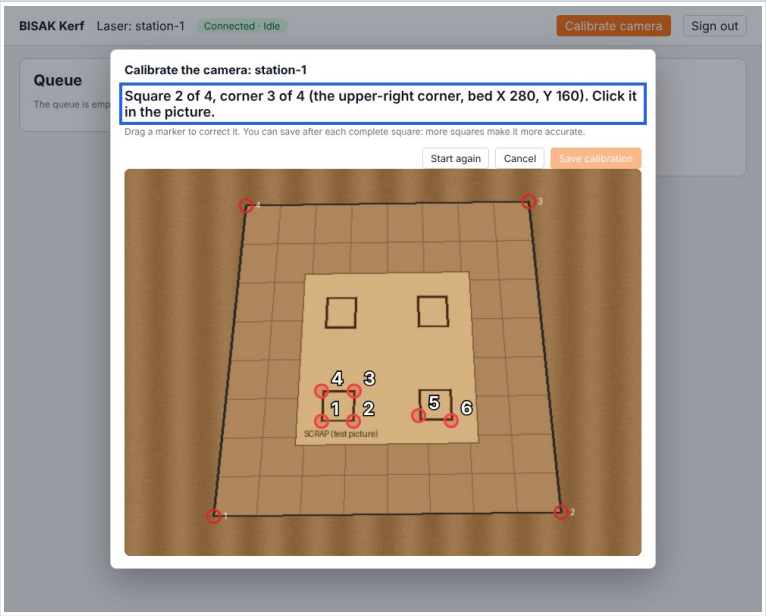
6 Click the first corner



When the cut is done, the live picture shows the four squares. The prompt tells you exactly which corner to click next, and its position on the bed. Do not move the material.

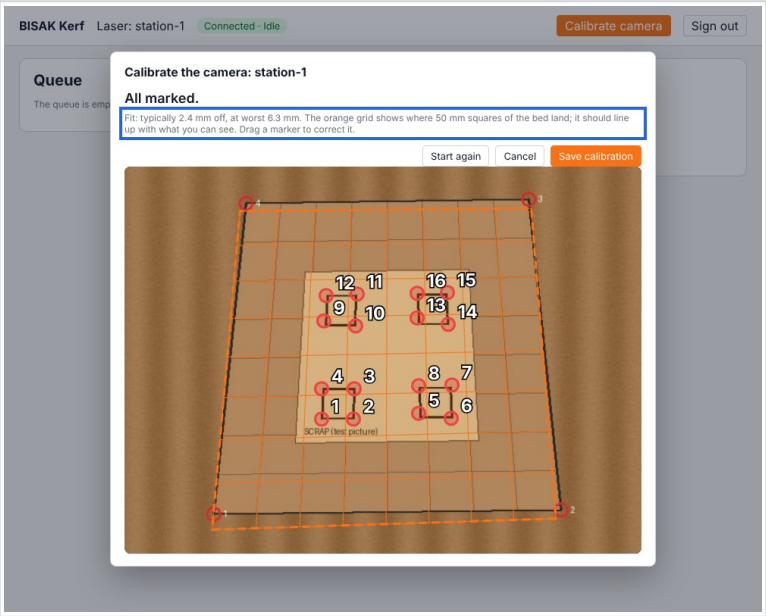
- For every square, click the corners in this order: **lower-left, lower-right, upper-right, upper-left** (counter-clockwise, as the bed's X and Y axes run).
- Click the outer corner of the scored line.

7 Keep clicking, square by square



Each click drops a numbered marker. Three corners of the second square are already placed here. If a marker is a little off, **drag it** to the right spot. Four squares means 16 clicks.

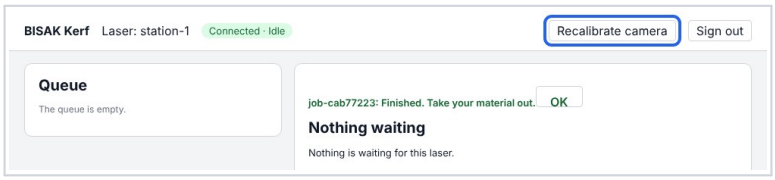
8 Check the fit



When every corner is marked, the dialog shows how well the points fit together: **typically** and **at worst**, in millimetres. The **orange grid** is where 50 mm squares of the bed land in the picture, and the dashed orange line is the bed's edge. They should sit on what you can see.

- A few millimetres typical error is normal for clicking by hand on a camera picture. If one number is much larger than the rest, drag that marker or press **Start again**.
- If the dialog says the points do not fit together, a square has been clicked in the wrong order or a click is far off. The picture below the message will show which marker is wrong.
- You can save after any complete square, but more squares make it more accurate.

9 Save

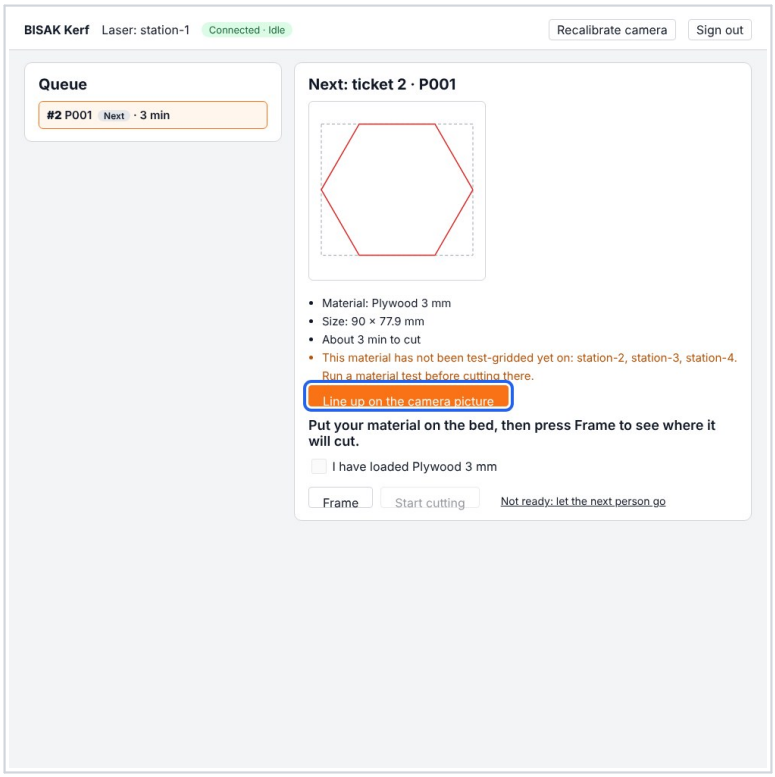


Press **Save calibration**. The Station page now offers **Recalibrate camera**, and the Admin site's Lasers page shows the laser as **camera calibrated**. Until a laser is calibrated, pupils at that laser can still move the head to their material and press Frame, as before.

Part 2 · What pupils then see

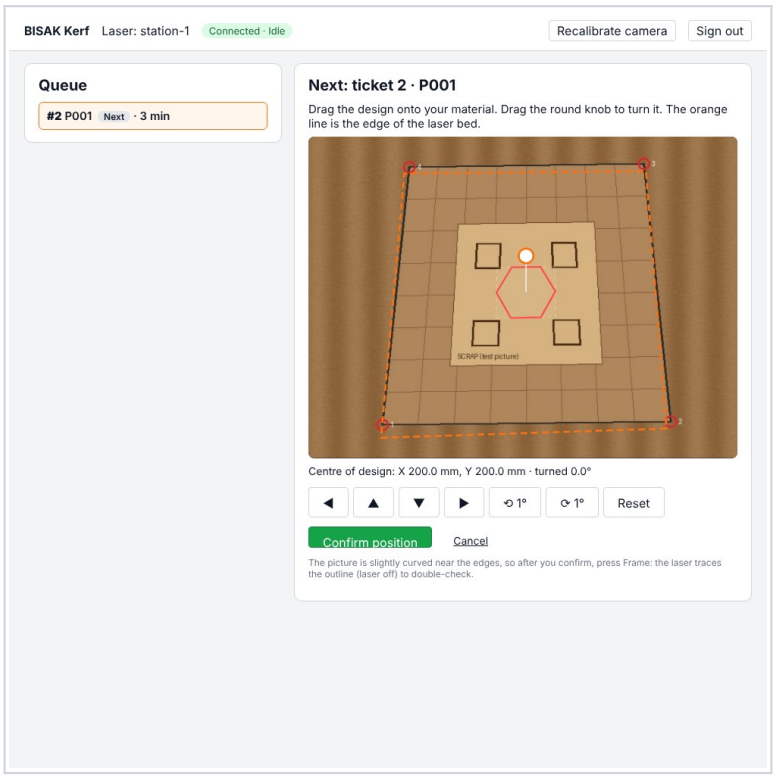
Once the laser is calibrated, the next job in the queue can be lined up on the camera picture.

10 Line up on the camera picture



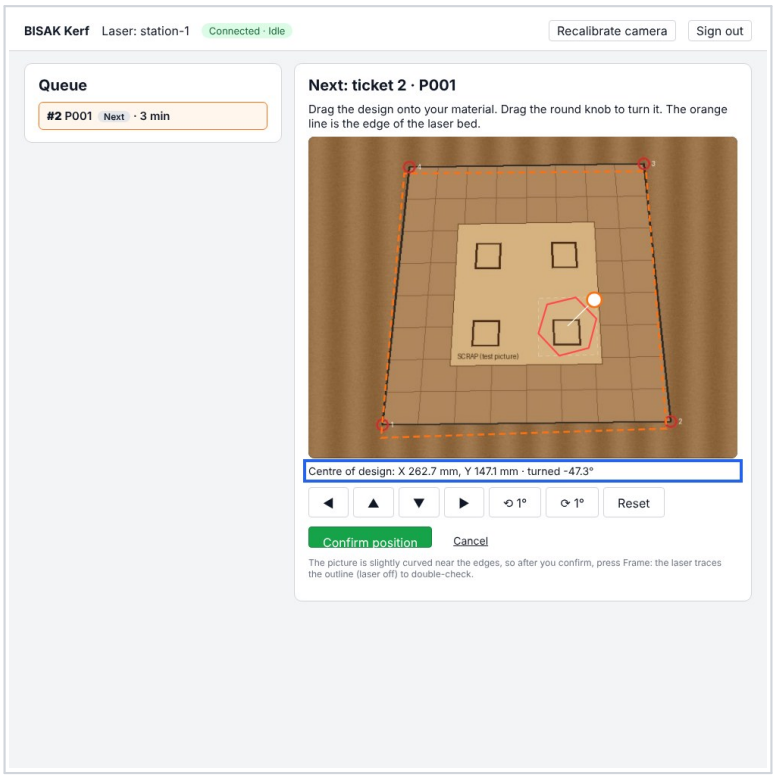
The pupil whose file is next sees its outline and details as usual, plus a new **Line up on the camera picture** button.

11 The design floats over the live picture



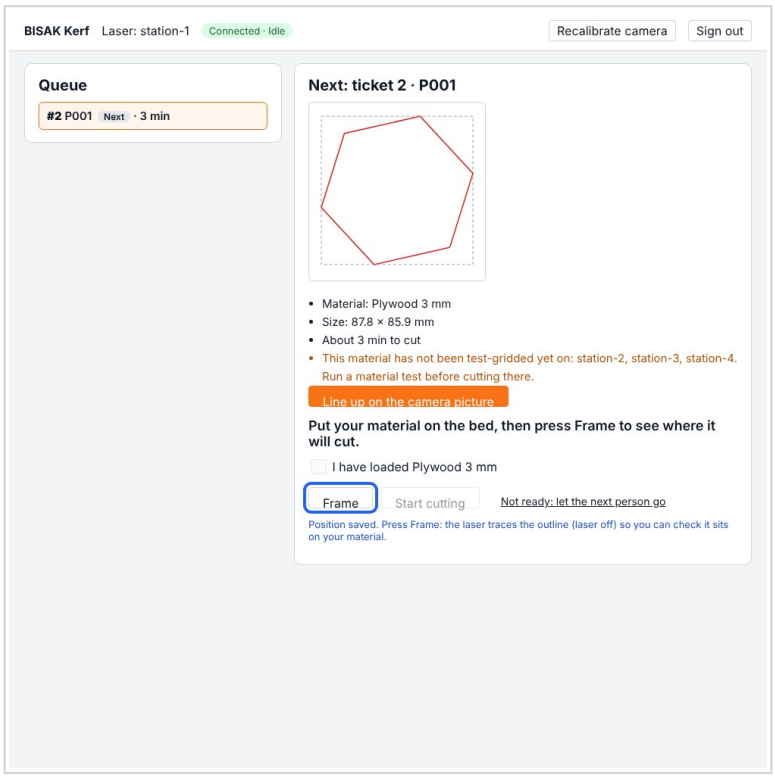
The red outline is the design. The orange dashed line is the bed. The pupil drags the design onto their material and drags the round white knob to turn it. The arrow buttons nudge it 1 mm and the turn buttons turn it 1 degree.

12 Move and turn it into place



The line under the picture shows where the design's centre is on the bed and how far it is turned. The **Confirm position** button is disabled while any part of the design is off the bed, with a message saying so.

13 Confirm, then Frame to double-check



Confirm position fixes the design at exact bed coordinates. The pupil then presses **Frame**: the laser traces the outline with the laser off so they can see it sits on the material. Only after that can they tick that the material is loaded and press Start.

Tips and troubleshooting

Getting a good calibration

- **Spread the squares out.** The map is accurate inside the area the marks cover and only extrapolates outside it. Use the widest spacing the camera can see.
- **Do not move the scrap** between cutting and clicking.
- **Use Score** for clean, high-contrast lines that do not cut the scrap into pieces.
- **Click the same corner each time**: the outer corner of the line, not the middle of it.

Know the limits

- This is a flat map of the picture onto the bed. It does **not** correct the camera's fisheye curve, so it is least accurate near the edges of the picture. That is why **Frame** stays as the real check.
- A more accurate method (lens calibration) can replace the stored map later without changing how anyone uses it.

When to calibrate again

- The camera was moved, bumped or replaced, or the camera's resolution changed. Pupils see a warning if the picture size no longer matches.
- Designs consistently land a few millimetres from where the pupil put them, even after Frame.

If something does not work

- **No Calibrate camera button:** no camera picture is arriving from that laser PC. Check the Cutter is running and its camera is working.
- **"That is not a teacher's token":** use a teacher or admin token, not a pupil or station token.
- **The laser is busy / not connected:** wait until the Station page says Connected · Idle.
- **The squares are off the bed:** change the middle X and Y, or reduce the spacing, so every square fits inside the bed.
- **To remove a calibration:** open Recalibrate camera, then press Remove calibration.